

Solve Electrical Circuits With Solve Elec

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What a newcomer to designing electrical circuits dreads is usually the mathematics behind it. Complicated equations expressing complex behaviour, calculations that do not tally, awe-introducing patterns and intriguing measures that send shocks down the spine. You as a designer could do with some help, would you not agree?

Solve Elec, as the name says, is a software that makes the task of dealing with electrical designs a lot easier. Right from the circuit stage, analysis, to finally getting a report, Solve Elec is your friend to help and correct the multitude of ideas developing in your mind. Read on to know what Solve Elec has to offer to make designing easier and fun.

An all-in-one workstation

It is often comfortable to have the various elements of the design stage on one screen, right next to each other. Especially in complex circuits where each click of the mouse leads to different scenarios, one needs to shift between the various pop-up windows to figure out what is happening where.

The workstation of Solve Elec is designed to hold the basic necessary tabs, neatly arranged in columns on a single screen. Be it the design tools for creating, running and editing a circuit, a window showing its properties or the measurement setup that helps you monitor your circuit—each has its own space that can be customised. Easy to see, easy to access!

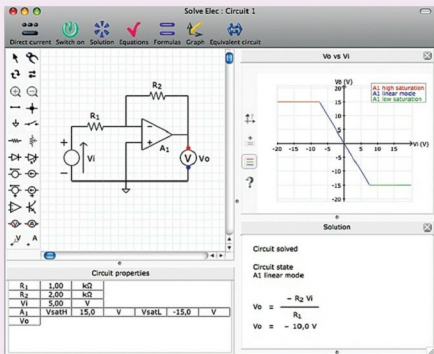


Fig. 1: A circuit on Solve Elec

Modes that replicate current behaviour

Right at the start, mode of operation has to be selected. This turns out to be the most important step as not all components can be ported between alternating current (AC) and direct current (DC) modes, which only seems logical after all.

In DC mode, components from switches to integrated amplifiers are available to you, while voltage and intensity sources drive a constant value for the entire duration of the simulation.

In AC mode, voltages, intensities and potentials are sine waves of the same frequency; the waves represented as complex values in circuit properties. Methods of com-

plex analysis are exploited to find a solution for the circuit that can only contain components operating in linear mode, so that the sine wave is preserved.

Features aiding effective analysis

A graph is one of the easiest means of analysing relationships between data, and the graph plotter in Solve Elec lets you plot quantities associated with components defined by metres or by formulae. The customisable graph asks you to define a parameter from among the component-associated quantities, while plotting metre or formula values on both axes.

There are also curve functions that help draw intensity and tension